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#define BLYNK_TEMPLATE_ID "TMPL3uJXE43xc"
#define BLYNK_TEMPLATE_NAME "weather"
#define BLYNK_AUTH_TOKEN "sH4tpASBtrraaKt3Ouxxq0CJvKhUlLjk"

char ssid[] = "esp";
char pass[] = "esp12345";

#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>
#include <ESP8266WebServer.h>
#include <ESP8266mDNS.h>
#include <WebSocketsServer.h>
#include <Wire.h>
#include <Adafruit_GFX.h>
#include <Adafruit_SSD1306.h>
#include <DHT.h>

#define DHTPIN D4
#define DHTTYPE DHT11
#define OLED_ADDR 0x3C
#define UPDATE_INTERVAL 1000

DHT dht(DHTPIN, DHTTYPE);
Adafruit_SSD1306 display(128, 64, &Wire);
ESP8266WebServer server(80);
WebSocketsServer websocket(81);      // websocket on port 81
BlynkTimer timer;

float t = 0, h = 0;
bool mdnsStarted = false;

// ---- Web page with Chart.js ----
const char HTML_PAGE[] PROGMEM = R"rawliteral(
<!DOCTYPE html><html><head>
<meta charset='UTF-8'>
<meta name='viewport' content='width=device-width,initial-scale=1'>

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<title>Weather Station</title>
<script src='https://cdn.jsdelivr.net/npm/chart.js'></script>
</head><body style='font-family:sans-serif;text-align:center'>
<h2>Weather Station</h2>
<canvas id='chart' width='350' height='200'></canvas>
<p>Temp: <span id='temp'>--</span> °C &nbsp; | &nbsp; Hum: <span
id='hum'>--</span> %</p>
<script>
const ws = new WebSocket('ws://' + location.hostname + ':81/');
let chart = new Chart(document.getElementById('chart').
getContext('2d'), {
  type:'line',
  data:{labels:[],datasets:[
    {label:'Temp °C',borderColor:'red',data:[],yAxisID:'y1'},
    {label:'Hum %',borderColor:'blue',data:[],yAxisID:'y2'}
  ]},
  options:{animation:false,scales:{y1:{type:'linear',position:'left'},
y2:{type:'linear',position:'right'}}}
});
ws.onmessage = e=>{
  let d = JSON.parse(e.data);
  document.getElementById('temp').textContent=d.t.toFixed(1);
  document.getElementById('hum').textContent=d.h.toFixed(1);
  let time = new Date().toLocaleTimeString();
  chart.data.labels.push(time);
  chart.data.datasets[0].data.push(d.t);
  chart.data.datasets[1].data.push(d.h);
  if(chart.data.labels.length>20){
    chart.data.labels.shift();
    chart.data.datasets[0].data.shift();
    chart.data.datasets[1].data.shift();
  }
  chart.update();
};
</script></body></html>
)rawliteral";

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// ---- Handlers ----
void handleRoot() {
    server.sendHeader("Content-Type", "text/html; charset=utf-8");
    server.send_P(200, "text/html", HTML_PAGE);
}

void handleJson() {
    String json = "{\"t\":\"" + String(t,1) + "\",\"h\":\"" + String(h,1) +
    "\"}";
    server.send(200, "application/json", json);
}

// ---- Core logic ----
void readSensor() {
    float newH = dht.readHumidity();
    float newT = dht.readTemperature();
    if (!isnan(newH) && !isnan(newT)) {
        h = newH; t = newT;
        Blynk.virtualWrite(V0, t);
        Blynk.virtualWrite(V1, h);
        // push to websocket clients
        String msg = "{\"t\":\"" + String(t,1) + "\",\"h\":\"" + String(h,1)
+ "\"}";
        websocket.broadcastTXT(msg);
    }
}

void showOLED() {
    display.clearDisplay();
    display.setTextColor(SSD1306_WHITE);
    display.setTextSize(1);
    display.setCursor(0, 0);
    display.print("Weather Station");
    display.drawLine(0, 10, 127, 10, SSD1306_WHITE);
    display.setTextSize(2);
    display.setCursor(0, 20);
    display.printf("%.1fC", t);
}

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display.setTextSize(1);
display.setCursor(0, 48);
display.printf("Humidity: %.0f%%", h);
display.display();
}

void setup() {
  Serial.begin(115200);
  Wire.begin(D2, D1);
  display.begin(SSD1306_SWITCHCAPVCC, OLED_ADDR);
  dht.begin();

  WiFi.begin(ssid, pass);
  Serial.print("Connecting to WiFi");
  while (WiFi.status() != WL_CONNECTED) { delay(500);
Serial.print("."); }
  Serial.println("\nConnected, IP: " + WiFi.localIP().toString());

  Blynk.config(BLYNK_AUTH_TOKEN);
  Blynk.connect();

  server.on("/", handleRoot);
  server.on("/json", handleJson);
  server.begin();
  websocket.begin();
  websocket.onEvent([](uint8_t, WStype_t, uint8_t*, size_t){});

  if (MDNS.begin("weather")) {
    mdnsStarted = true;
    MDNS.addService("http", "tcp", 80);
    MDNS.addService("ws", "tcp", 81);
    Serial.println("mDNS running: http://weather.local");
  }

  timer.setInterval(UPDATE_INTERVAL, []() {
    readSensor();
    showOLED();
  });
}

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    });  
}  
  
void loop() {  
  Blynk.run();  
  timer.run();  
  server.handleClient();  
  websocket.loop();  
  if (mdnsStarted) MDNS.update();  
}
```